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Extend the Camera Trap Data Package standard for bioacoustics data

InsectAI datathon

RESEARCH INSTITUTE NATURE AND FOREST

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Project: Safe & sound

- Review of existing standards and guidelines for bioacoustics
- Reach out to users
- Collect use cases

Conclusions

- Need for a widely accepted bioacoustics data standard
- Camtrap DP fits and can be extended

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arm

NINA
Norwegian Institute for Nature Research

Bioacoustic changes to Camtrap DP

Suggested changes

- 37 suggestions discussed
- Logged on Github: https://github.com/tdwg/camtrap-dp/milestone/4

Suggested changes: examples

1) **Change names**
   - cameraModel → **device**Model

2) **Extend descriptions**
   - captureMethod
     - activityDetection
     - timeLapse
     - continuous

Suggested changes: examples

3) **Add new terms**

- samplingFrequency
- gain
- channels

4) **Add table**

- assertions (measurements or facts)

Metadata

#	Term	Suggested by	Decision	Rationale
P1	coreRealm	WWSS	wontfix	In what realm the recording took place (e.g. “freshwater”, “marine”, “terrestrial”, “subterranean”). This can be expressed in keywords (e.g. “freshwater” or “realm:freshwater”).
P2	medium	WWSS	wontfix	In what medium the recording took place (e.g. “soil”, “water”, “air”). This can be expressed in keywords (e.g. “soil” or “medium:soil”).
P3	protocolType	INBO	add	Recording type (e.g. “camera-trapping”, “bioacoustics”) as a high-level differentiation between datasets. Could contain multiple values, cf. observationLevel.
P4	classificationEffort	Cornell	add	To what extent the media have been classified. While a percentage or controlled vocabulary might be useful, free text will better capture nuances (e.g. “all mammals except rodents”).
P5	captureMethod	NINA	update	Some devices can record continuously. This can be expressed by extending the controlled vocabulary for captureMethod to “continuous” (in addition to “activityDetection” and “timeLapse”).

Deployments (D1-D5)

#	Term	Suggested by	Decision	Rationale
D1	elevation	WWSS, Cornell	add	Elevation in meters at the location. May be derived from the latitude/longitude, but direct recording would be preferred.
D2	recordingSchedule	Humboldt Institute, Cornell	add	Free text field that describes when a device was recording. While a controlled vocabulary might be useful, free text will probably better capture nuances (e.g. “continuous from 9PM to 6AM local time and activity detection during the day). This term is similar to Movebank’s dutyCycle, but the name recordingSchedule is preferred. Alternatively, this information is recorded in deploymentComments.
D3	microphone	Humboldt Institute, WWSS	wontfix	The name/model of the microphone (i.e. physical sensor), which can be different for the same recording device (i.e. recorder). This can be expressed in cameraModel.
D4	targetTaxa	WWSS	unclear	The taxa that were the target of this study. This is different from taxonomic, which describes all taxa observed (which may or may not contain target taxa). Consider adopting Humboldt Extension terms at project or deployment level. For now, this information can be expressed as free text in project description.
D5	devicePlatform	DASA, Cornell	add	Substrate on which the device/camera is mounted. This information is especially useful for moving substrates and can have an effect on sound propagation. Best would be to use a controlled vocabulary, such as “buoy”, “vegetation/tree”, “building/wall”, “structure”, “unattached”. There are also moving platforms, such as “vehicle/boat”, “tow”, “animal”, “human”, but this would require a separate deployment for every location (Camtrap DP is designed for fixed platforms).

Deployments (D6-D8)

#	Term	Suggested by	Decision	Rationale
D6	cameraHeight	INBO	update	Devices can be subterranean (e.g. in a burrow). This can be expressed as negative values in cameraHeight, but its definition and constraints should be updated. Note that this term is already aligned with dwc: minimumDistanceAboveSurfaceInMeters, which uses negative values in the same way: “Use positive values for locations above the surface, negative values for locations below.” It is therefore a better match than cameraDepth, which should be reserved for underwater devices.
D7	deviceID, deviceModel, deviceDelay, deviceHeight, deviceDepth, deviceTilt, deviceHeading, deviceSetupType	Humboldt Institute	rename	“Camera” is not a term used for bioacoustic records. Suggest to rename terms containing “camera” to broader “device”. This rename: - Affects 3 metadata descriptions - 7 deployment terms - 0 media terms - 3 observation term The name “Camtrap DP” should likely be retained as it is widely used.
D8	calibration	VLIZ	unclear	Calibration or sensitivity value(s) for the device. Cornell: allows to derive maximum detectionDistance. This term is not needed for model development. It might be better to provide assertion values, potentially with a calibration (TRUE/FALSE) at media level.

Deployments (D9-D10)

#	Term	Suggested by	Decision	Rationale
D9	locationType	Cornell	rename	featureType refers to a deployment characteristic (e.g. “roadPaved”, “burrow”), but the name makes it sound like it is data processing related. Suggest to rename this (e.g. “deploymentFeature”, “locationType”, “siteCharacteristic”).
D10	timestampIssues	NINA	delete	timestampIssues is a boolean flag. This may need to be removed for two reasons: 1. It’s oddly specific, there are probably better ways to implement quality flags. 2. In bioacoustics, the two most crucial information are the amplitude and the time. The guidelines should indicate that deployments with missing/incorrect timestamps that cannot be corrected should not be included.

Media (M1-M4)

#	Term	Suggested by	Decision	Rationale
M1	captureMethod	NINA	update	Some devices can record continuously. This can be expressed by extending the controlled vocabulary for captureMethod to “continuous” (in addition to “activityDetection” and “timeLapse”).
M2	captureMethod	INBO	update	Rename the controlled value “timeLapse” to “schedule” or “recordingSchedule”, which is more broadly understandable and in alignment with change D2.
M3	frequencyRange		wontfix	The maximum frequency range. This can be deduced from the samplingFrequency, e.g. if the sampling frequency is 32kHz, the maximum frequency range is 0-16kHz. See also O5.
M4	samplingFrequency	NINA, Humboldt Institute	add	Frequencies at which measurements were made, also called sampleRate. Expressed in hertz (preferred over kHz, so integers can be used), e.g. “44100”, with a minimum value of 0. While this is a device (and therefore deployment) property that is unlikely to change during the deployment, it can be reduced/compressed in the resulting media file. Since we want users to be able to assess if the media file is useful for them, we should record this at media-level. The same is true for M4, M5, M6 and M7. Might also be a useful project-level property. Equivalent to Sample Rate in Audiovisual Core. Can this also be a property for video?

Media (M5-M9)

#	Term	Suggested by	Decision	Rationale
M5	bitDepth	NINA, Humboldt Institute	add	Number of bits of information in an audio file (see Wikipedia). Expressed as a unitless integer with 4 distinct values: 8, 16, 32, 64. Bit depth can be reduced when exporting an audio file, but not recovered. Has an impact on AI model development.
M6	gain	NINA	add	Amplification of the signal (see Wikipedia). High gain results in more background noise. Expressed in numeric decibel (dB) with a minimum value of 0.
M7	channel	WWSS, VLIZ	add	Number of channels, from 1 (mono), 2 (stereo) up to 8. Expressed as a unitless integer with 8 distinct values. Can this also be a property for video?
M8	duration	DASA, Humboldt Institute	add	Duration of an audio and video file. Expressed as a numeric value, in seconds. Useful to know without having to open the file or to validate that observation eventStart and eventEnd fall within the timestamp+duration. Sometimes referred to as “fileLength”. Equivalent to Media Duration in Audiovisual Core, but note that this definition takes into account media speed.
M9	fileSize	DASA	wontfix	File size in bytes. Potentially useful to assess how long a file will take to download. Not a good alternative for duration (M8). No strong use cases.

Media (M10-M12)

#	Term	Suggested by	Decision	Rationale
M10	snippet	INBO	wontfix	Indication if the media file is a snippet/subset of a longer media file. Can largely be assessed from timestamp+duration. No strong use cases.
M11	timestamp	INBO	update	Support milliseconds in timestamp, as well as observations eventStart and eventEnd. Suggested as https://github.com/tdwg/camtrap-dp/issues/333
M12	channels	Cornell	wontfix	An array of hydrophones may result in multi-channel audio files, with each channel representing a single hydrophone. This (fairly rare) use case will not be supported and should be expressed as multiple deployments paired with deploymentGroups and more info (what channel is used, how channels are synchronized, etc.) expressed in deploymentComments or mediaComments.

Observations (O1-O4)

#	Term	Suggested by	Decision	Rationale
O1	observationLabels	NINA	wontfix	Rename observationTags to observationLabels. The word “label” is often used in software. Since it has the same meaning as tags, it’s best to not implement this change.
O2	detectionID	NINA	wontfix	A single media file can contain multiple observations/detections. This can currently be expressed by having multiple rows in observations, each referring to the same mediaID and with their own observationID, eventStart and eventEnd. No change is needed to support this use case.
O3	annotationMethod	NINA	wontfix	How the observation was created and with what probability. This can currently be expressed with classificationMethod (human/machine), classifiedBy (e.g. the name of the model) and classificationProbability. No change is needed to support this use case.
O4	classificationProbability	INBO	wontfix	Update the definition for classificationProbability to indicate that 100% for a classification by a human means that the classification was validated. Interpreted as such by the camtrapdp R package. Is likely better expressed in a new term, see O8.

Observations (O5-O7)

#	Term	Suggested by	Decision	Rationale
O5	frequencyLow, frequencyHigh	NINA	add	The frequency range to which the observation applies, similar to the temporal range defined by eventStart and eventEnd or spatial range by the bbox terms. Note that bbox terms could be used for spectrograms, but those are not the media files that are shared in media. Camtrap DP currently doesn't support sharing multiple versions of the same media file (e.g. raw file, compressed file, spectogram). Equivalent to Lower frequency bound and Upper frequency bound in Audiovisual Core.
O6	observationLevel	INBO	update	Observations can refer to a subset of a media file. This can be expressed by extending the controlled vocabulary for observationLevel to "interval" or "subset" (in addition to "event" and "media"). Suggested as https://github.com/tdwg/camtrap-dp/issues/390
O7	latitude, longitude	INBO	wontfix	Camtrap DP currently does not support moving devices. To record this, one needs to split the deployments into one row per location. Another option would be to indicate a more precise latitude and longitude in the observations table. Since not all locations might have an observation, it would lead to an incomplete track. Adding latitude and longitude to the media table won't work, since media files can have a duration during which the device was moved.

Observations (O8-O11)

#	Term	Suggested by	Decision	Rationale
O8	classificationConfirmation	DASA, UCL	add	A confirmation that the classified species was observed via other means (e.g. “captured”, “heard on site”, “seen on site”). Could be named “reliabilityType” or “classificationConfirmation”. It would help AI model development and is a type of certainty that is different from 100% classificationProbability, which is solely based on the available media file. Could be expressed in observationTags, but a separate term with controlled vocabulary would be helpful. Could be used to record “validated” (O4).
O9	vocalActivityLevel	Humboldt Institute	wontfix	A number indicating how active the species, from a single individual to a chorus. Numbers might be a range (e.g. “0-5”). Likely better expressed as count (no ranges) or observationTags.
O10	(table definition)		add	Update the definition of the observations table to indicate that it only contains the consensus across all classifications/annotations of a media file or event. I.e. it does not contain conflicting information.
O11	behavior	Tom August	update	Update the examples in behavior to include behavior that can be deduced from sound (e.g. alarm call, territorial call, echolocation call, feeding buzz).

Assertions: 3 solutions

#	Suggestion	Suggested by	Decision	Rationale
A1	assertions (new table)	INBO, NINA	unclear	Add a table to capture extra information about deployments, media and observations. Similar to measurementOrFacts in Darwin Core Archive, **each row represents an assertion** (with type, value, unit, reference) and is linked to the respective table via a foreign key: deploymentID, mediaID or observationID (all three fields are listed, only one should be populated). Compared to A2, only one table is needed. Suggested as https://github.com/tdwg/camtrap-dp/issues/358
A2	deploymentAssertions, mediaAssertions, observationAssertions (3 new tables)	INBO, NINA	unclear	Add three tables to capture extra information about deployments, media and observations (one for each). Similar to assertions in Darwin Core Data Package, **each row represents an assertion** (with type, value, unit, reference) and is linked to the respective table via a foreign key: deploymentID, mediaID or observationID (only one field is listed). Compared to A1, it allows stricter validation and uses the same approach as in Darwin Core Data Package.
A3	Custom fields	INBO	unclear	Allow publishers to add extra fields to the deployments, media, and observations tables, where **each column represents an assertion**. Compared to A1 and A2, it is a lot easier for publishers, but only allows to express values (not units or a reference). It would require the fieldMatch property and therefore an upgrade to Data Package v2. A side benefit of this would be that non-required terms do not need to be provided.

Assertions

Examples of how such a table could look like:

- https://rs.gbif.org/extension/dwc/measurements_or_facts_2025-07-10.xml
- https://github.com/gbif/dwc-dp/blob/master/dwc-dp/table-schemas/event-assertion.json
-

Next steps

Next steps

- Start TDWG (Biodiversity Information Standards) Task Group
- Continue the discussion in Github issues[1]
- Prepare the bioacoustic version of Camtrap DP

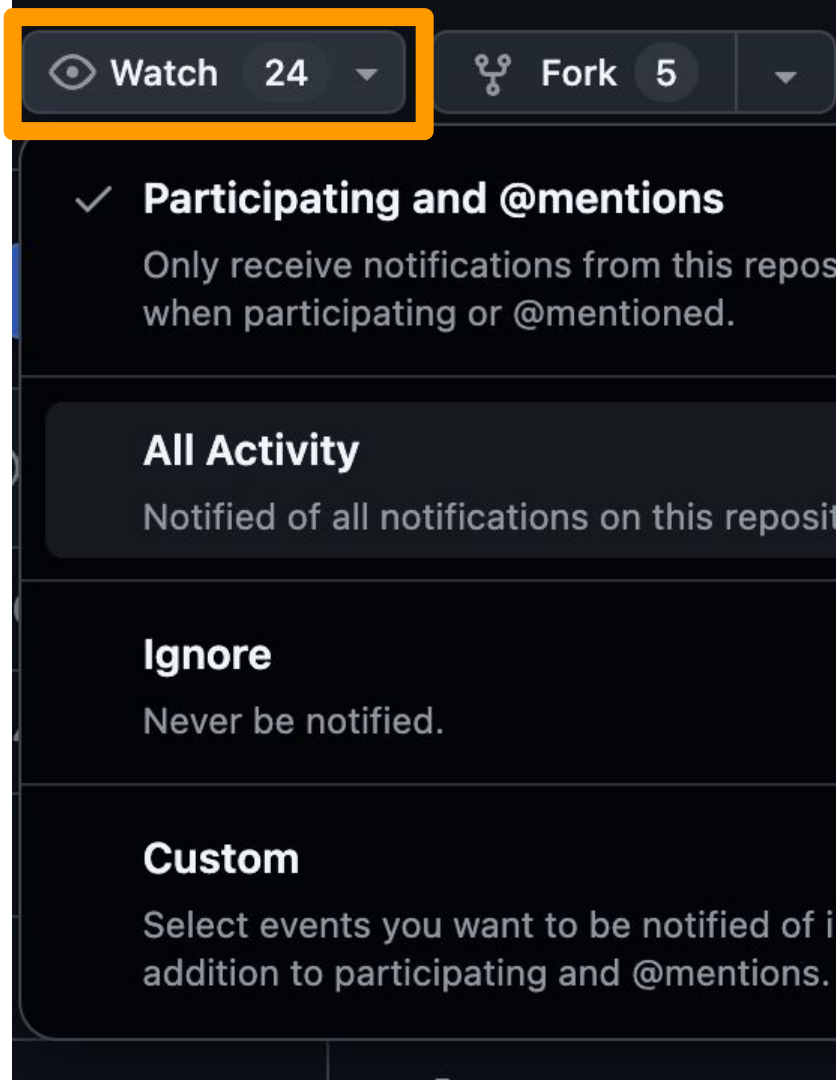
Release of a new version of the standard is dependent on:

- How fast we get to consensus
- Funding

[1] https://github.com/tdwg/camtrap-dp/issues

How to be involved?

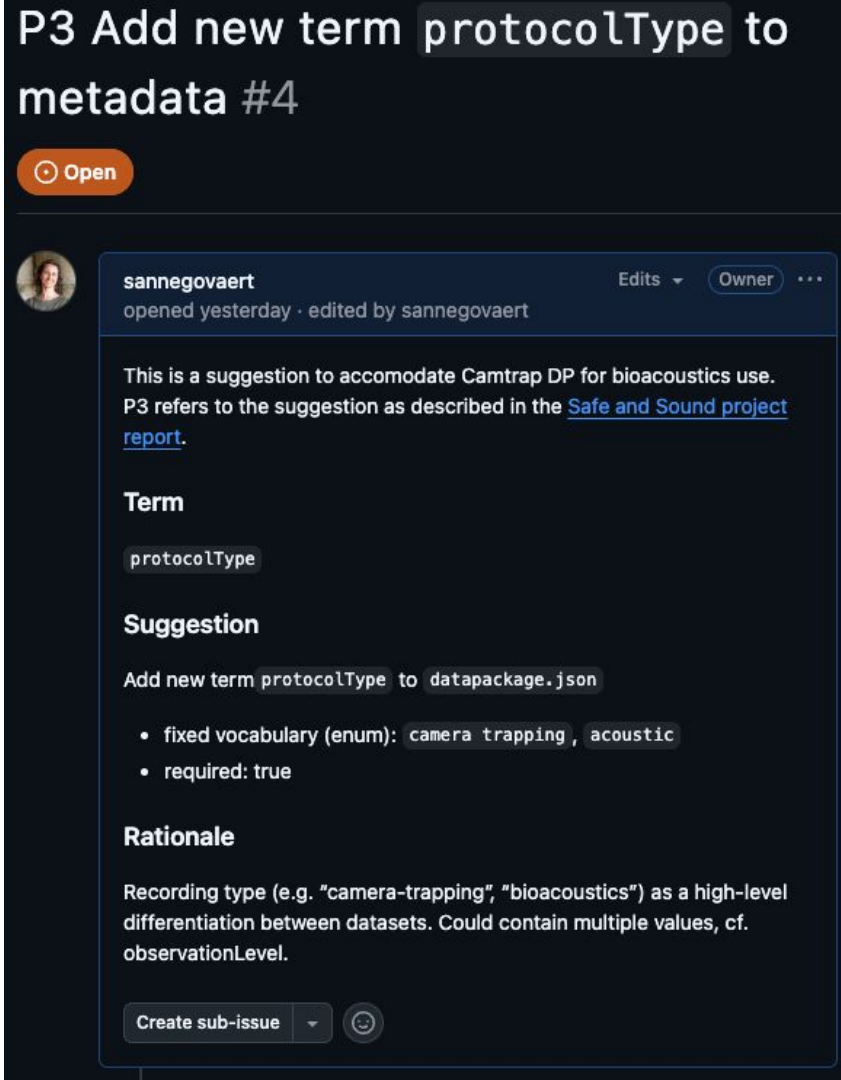
1) Keep **updated:**
   - watch the Camtrap DP GitHub repo[1] to get notifications
   - Read the report[2]



[1] https://github.com/tdwg/camtrap-dp
[2] https://hdl.handle.net/11250/5343743
[3] https://wildlabs.net/discussion/safe-and-sound-standard-bioacoustic-data

How to be involved?

2) **Contribute** to the discussion in [GitHub issues](https://github.com/tdwg/camtrap-dp/issues)[1]



[1] https://github.com/tdwg/camtrap-dp/issues

How to be involved?

3) **Test** your data[1]

[1] https://github.com/camera-traps/bioacoustics

How to be involved?

More specifically…

- ➔ Be an early adopter
- ➔ Share funding opportunity
- ➔ Help us to connect with ecoacousticians
- ➔ Become member of the Task Group (soon…)

Questions or comments? Contact us by mail!
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Thank you!

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